Presentation script

8-minute talk by Kiet, Hieu and Dung; Phong runs the live dashboard demo after the talk. The same text is in the speaker notes of retention-presentation.pptx.

Speaker	Slides	Time
Kiet	1-5	2:40
Hieu	6-9	2:40
Dung	10-13, 15-16	2:40
Phong	14	demo, not timed
Total		**8:00**

Slide 1: Title

Kiet | 0:00-0:15 | 15 s

Good morning, Professor, and everyone. We are Group 1, presenting Topic 7: profit-aware telecom churn and retention targeting. Hieu, Dung and I will present, and Phong will demo our dashboard at the end.

Slide 2: Outline

Kiet | 0:15-0:25 | 10 s

We follow CRISP-DM: the business problem and the data, then the model and the decision rule, then results and robustness, and finally our recommendation.

Slide 3: The Decision: Who Gets a Retention Offer?

Kiet | 0:25-1:15 | 50 s

In our sample company 26.5% of customers left, which puts about $468,468 of twelve-month margin at risk. The retention team has a fixed budget, and today it calls the customers with the highest churn risk. But every offer costs money, and a risky customer with a small bill may not be worth saving. So our question is: with a limited budget, who should receive an offer when both risk and value count? Our answer is to score each customer and offer by expected value: churn probability, times the save rate, the share of would-be leavers an offer keeps, times customer value, minus the cost. We contact a customer only when it is positive.

Slide 4: Data and Validation Design

Kiet | 1:15-2:00 | 45 s

We use the IBM Telco Customer Churn sample from Kaggle: 7,043 customers of a fictional US operator. The unit of analysis is the customer, the target is churn, and the decision is whether to make an offer, and which one, within the budget. Quality checks found no duplicates and only 11 blank total charges, all for brand-new customers, which we set to zero. The data is one snapshot without dates, so we used a stratified 60/20/20 split: every choice is made on validation, and the test set is used once. To avoid leakage we excluded churn reasons and IBM's churn score, which are known only after a customer leaves.

Slide 5: Key Insight: Churn Risk Is Not Customer Value

Kiet | 2:00-2:40 | 40 s

The exploratory analysis gave us our key insight. Churn is highest for month-to-month contracts, new customers, fibre internet and electronic-check payers. But risk and value only partly overlap. The high-risk, low-value group holds 46% of the value at risk, yet each of these customers is worth little. Only 57 test customers are both high-risk and high-value, and today's risk-only rule reaches just 12% of them. So ranking by risk alone spends the budget in the wrong place. Hieu will now show how we fix that.

Slide 6: Outline

Hieu | 2:40-2:45 | 5 s

Thank you, Kiet. I will start with the churn model.

Slide 7: Churn Model: Simple, Calibrated, Good Enough

Hieu | 2:45-3:35 | 50 s

We compared three model families: logistic regression as the baseline, a random forest as the required tree model, and XGBoost. Each was tuned with five-fold cross-validation and calibrated. On the untouched test set they rank customers almost equally well, so, by a rule we fixed in advance, we deploy the simplest: calibrated logistic regression. Its ROC-AUC is 0.83 and its PR-AUC 0.62, against 0.27 for a model that knows nothing. Calibration is essential here, because we multiply the probability by money: the calibration error is only 0.019. The strongest drivers are month-to-month contracts, which multiply the odds of churn by about 3.7, and fibre internet; longer tenure lowers the risk.

Slide 8: From Probability to Offer: Expected Value

Hieu | 3:35-4:30 | 55 s

Next we turn a probability into a decision. The expected value of an offer is the churn probability, times the save rate, times the customer's value, minus the offer and contact cost. Customer value is the monthly bill times a 30% margin over a discounted twelve-month horizon, longer for loyal customers and shorter for new ones. We defined a small catalogue of offers, for example a $35 credit for moving to a one-year contract, or six months of free tech support. An integer program then chooses which customers get which offer to maximise total expected value within the budget, with one offer per customer, eligibility rules and a contact cap. The save rates are assumptions, not measurements, and we treat them that way.

Slide 9: Same Budget, $1,368 More Expected Value

Hieu | 4:30-5:20 | 50 s

Here is the result on the 1,409 test customers, all with the same $3,500 budget. Calling random customers, or the most valuable ones, loses money, because most of them would have stayed anyway. Today's highest-risk rule earns about $1,301. Expected-value targeting with segment offers earns $2,668: that is $1,368 more for the same spend, $1,039 from choosing better customers and $328 from matching offers. When we revalue the lists with customers' actual churn, the gain is still positive, with a 95% interval from $229 to $2,002. Dung will now explain how much to spend and how sure we are.

Slide 10: Outline

Dung | 5:20-5:25 | 5 s

Thanks, Hieu. Let me turn to the budget and to risk.

Slide 11: How Much to Spend: The Profit Curve

Dung | 5:25-6:05 | 40 s

How big should the budget be? Each extra step of budget returns less. Our rule, fixed on validation data, keeps adding budget only while the last step still returns at least 10%. That gives enough budget for the standard offer to 26% of customers. On the test customers this buys 324 offers for $9,150, with an expected net value of $4,145. For the whole base of 7,043 customers that is about $45,737, or $6.49 per customer. The rule stops a little before the point where no customer has a positive expected value, because the last steps barely pay for themselves.

Slide 12: How Sure Are We?

Dung | 6:05-6:55 | 50 s

How sure are we? We varied every assumption one at a time, and then all of them at once in 5,000 Monte Carlo draws. In 99.96% of the draws, expected-value targeting beats the risk-only rule: who to target is robust. How much an offer saves is not. The recommended list breaks even if the standard offer saves 13.8% of would-be leavers, while we assumed 20%, and it loses money in 17% of the draws. IBM's churn reasons add a warning: only 46% of churners left for price or a competitor, which an offer can address. Finally, our fairness audit shows that seniors are contacted 3.3 times as often as other customers, partly because of higher risk and partly because of higher bills, so every wave is reviewed.

Slide 13: Recommendation: A Randomised Pilot Before Scaling

Dung | 6:55-7:45 | 50 s

So our recommendation to the Head of Retention is to adopt expected-value targeting with segment offers, but to launch it as a randomised pilot. One wave of 1,620 customers from the recommended list is split at random: half receive the offer, half are held out. After 90 days we measure the real save rate of this offer mix, which we assume to

be 22.6%. The decision rule is agreed now: go if the effect is significant and at least the break-even of 15.6%; stop if it is clearly below. The offered half costs about $22,875, and if our assumption holds, the rule says go 97% of the time. A decision is ready about 19 weeks after approval. Phong will now show our dashboard.

Slide 14: Live Demo: Retention Lab Dashboard

Phong | live demo before the closing words, not part of the 8-minute talk | full timed steps in demo-script.md

Short version of reports/presentation/demo-script.md (Streamlit app, default assumptions, held-out test customers):

1. Decision page: read the five KPIs (expected net value $2,668, ROI 76%, 153 customers contacted, precision, high-risk/high-value coverage) and the green recommendation sentence.
2. Scroll to 'Who gets an offer': fibre, month-to-month and bill profile, then the risk x value scatter.
3. Set the budget to 0 and back to $3,500: KPIs and the action list update at once.
4. Action list: search one customer ID, open 'Why is a customer in or out of the list?', then click 'Export campaign' (CSV + assumptions).
5. Assumptions & sensitivity: lower the save rate to 10%; the plan shrinks and the live chance of a loss rises; point at the break-even save rate.
6. Model evidence: performance table and points scorecard; upload data/examples/customers_unlabeled.csv to score new customers.

Hand back: 'That is the tool a retention manager would use for every wave. Back to Dung.'

Slide 15: Team Contributions

Dung | 7:45-7:55 | 10 s

Who did what: Kiet led the project and the report, Hieu the decision engine, Phong the data and models, and I these slides.

Slide 16: Thank you

Dung | 7:55-8:00 | 5 s

Thank you for your attention. We are happy to take your questions.

Slide 17: Closing

No speaker

Closing slide: leave it on screen during questions.